LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic concept of ESM
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic concept of ESM. Types of Information extracted like PRF, PRT, and PW etc. and their utilization for verifying types of Radar.
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of ESM parameter definitions and Radar type identification logic based on extracted data; Safety: No live equipment handling; Instructor supervision: Monitor understanding of PRF/PRT/PW relationships.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an Electronic Support Measures (ESM) system?
(b) How does an ESM system differ from an Electronic Counter Measures (ECM) system?
(c) Why is the identification of radar types critical in Anti-Submarine Warfare (ASW) operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and Basic Concept of ESM • Role of ESM in ASW Operations • Function of Secondary Radars in ASW • Signal Detection and Processing Basics • Importance of Radar Identification	L & Ds	PPT, IB	15 min
(b)	• Pulse Repetition Frequency (PRF) Analysis • Pulse Repetition Time (PRT) Calculation • Pulse Width (PW) Measurement • Interpreting PRF/PRT/PW for Radar Type • Verification of Radar Types using Parameters	L & Ds	PPT, PDF	20 min
(c)	• Utilization of Extracted Data for Verification • Correlating Parameters with Radar Classifications • Limitations of ESM Data in ASW • Integration with Other Sensor Data • Summary of Radar Verification Techniques	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how PRF and PRT values are used to distinguish between search and fire-control radars.
(b) Describe the specific role of secondary radars in supporting ASW operations using ESM data.
(c) List three key parameters extracted by ESM and their utility in verifying radar types.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of ESM discussed in this lesson will assist in analyzing radar threats in the upcoming lesson on Electronic Counter Measures (ECM).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Basic concept of ECM. Principle of Jamming used as ECM aid.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min